Introduction to Economics

Class 14 — Monopolistic competition; problems on Blocks 3–4

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Plan for today

- Monopolistic competition: many sellers, each a little different
- A coworking space in the short run and the long run
- Why firms advertise
- Problems on Blocks 3–4: a tax; a price taker's costs
- An external cost; a price cap

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Block 4's last class. Class 12 had many sellers of an identical product and none of them able to move the price; class 13 had one seller that was the whole market. Most firms you buy from are neither: today's market has many sellers, each a little different from the others, and anyone free to open. The first half builds that case on one coworking space and ends with why such firms pay for adverts. The second half is practice: four problems on Blocks 3 and 4, worked in pairs, with the answer after each.

Slide 2 Plan for today

The class at a glance. Before the break, one idea built on one example: what is spent on adverts in Spain, which kind of market makes an advert worth paying for, a coworking space in the short run and after rivals open, and why it ends up with empty desks and a reason to advertise. After the break, four problems from Blocks 3 and 4 in pairs, each followed by its answer: a tax on a market, a price taker's cost table, an external cost, a price cap. Problems like these are good practice for Midterm 1 on Thursday 19 November.

Warm-up: a riding school's money flows

A riding school near a town

1. The field its horses graze, rented from a farming family: which factor, and its payment?
2. It builds new stables: a leakage or an injection?
3. It pays a tax on its profit: a leakage or an injection?

In pairs, two minutes: three answers.

Land, labour, capital: what households supply (class 3, slide 22) · some income leaks out, spending is injected back (class 3, slide 23).

Slide 3 Warm-up: a riding school's money flows

A warm-up on the circular flow, with a riding school near a town. Before class, name for each item which of the three factors is involved and what households are paid for it, and ask which flows leave the loop between firms and households and which put spending back in.

Revealed in class.

Warm-up: who should make the feeders?

Two workshops make birdhouses and bird feeders

hours for one	birdhouse	feeder
Fusta	2	1
Escaire	6	2

1. Who has the absolute advantage in both?
2. Who should make the feeders?
3. They swap 1 birdhouse for 2.5 feeders: do both gain?

In pairs, two minutes: three answers.

Comparative advantage: the lower opportunity cost (class 3, slide 18).

Slide 4 Warm-up: who should make the feeders?

A second warm-up, on comparative advantage, with two workshops that make birdhouses and bird feeders. Before class, work out from the table what one feeder costs each workshop in birdhouses given up, and ask which prices for a swap would leave both workshops better off.

Revealed in class.

Advertising in Spain: how much a year?

Revealed in class

Guess before class: how many euros a year are spent on advertising in Spain, per person living here? 2.60, 26, 260 or 2,600?

Slide 5 Advertising in Spain: how much a year?

A real number about something nearly every firm does: what is spent on advertising in Spain in a year. Before class, make your guess per person living in Spain, and ask which kind of firm would pay to win one more customer: one that sells all it wants at the market price, as in class 12, or one that sets its own price, as in class 13, and why.

Revealed in class.

Monopolistic competition: many sellers, each a little different, and anyone can open one

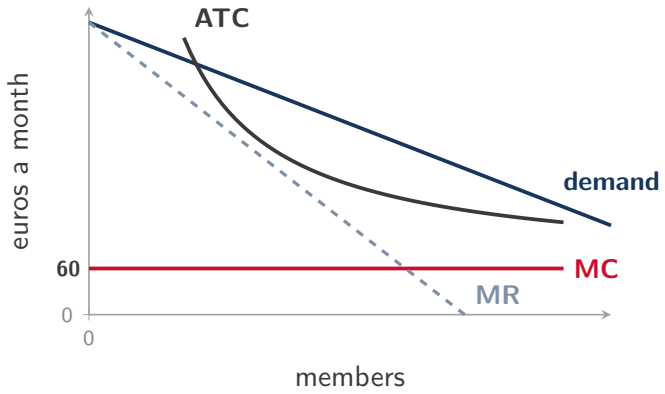
	Perfect competition (class 12)	**Monopolistic competition** (today)	**Monopoly** (class 13)
Sellers	many	many	one
Product	identical	each a little different	no close substitute
Entry	free	free	blocked
One firm's demand	flat at the price	slopes down	the whole market's

Price taker: sells all it wants at the market price (class 12, slide 6) · monopoly: the only seller, no close substitute (class 12, slide 20).

Slide 6 Monopolistic competition: many sellers, each a little different, and anyone can open one

The two outer columns are the last two classes; the middle column is today, and it borrows one feature from each. Monopolistic competition has many sellers, each selling a product that is at least a little different from the others (product differentiation: where it is, how it looks, how it is made, or simply how buyers see it), and free entry: anyone can open one or leave. The running example is coworking spaces in a city: each has its own location, light, coffee and people, and anyone can rent a floor and buy desks. Because its product differs, a space that raises its price loses some members but not all, so its own demand slopes down like a monopoly's: that is the "monopolistic" half. Because anyone can open, profits attract rivals, as under perfect competition: the "competition" half. The trap is to think that many sellers always means a price taker; that needs identical products as well.

A coworking space in the short run: how many members, at what price?



60 desks

rent, furniture and the owner's time: 3,600 a month

each member costs 60 a month

Marginal revenue (MR): what one more member adds to revenue (class 13, slide 4).

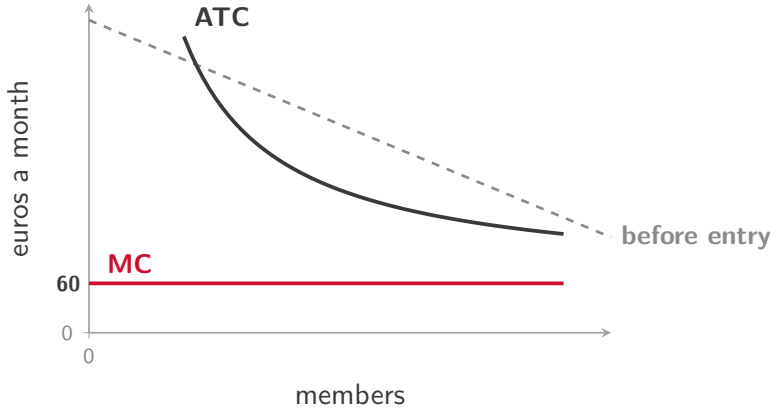
Average total cost (ATC): total cost ÷ members (class 11, slide 10) · MC: marginal cost.

Slide 7 A coworking space in the short run: how many members, at what price?

One typical space, with its numbers in the card: 60 desks, 3,600 euros a month for rent, furniture and the owner's time whatever happens, and 60 euros a month more for each member, which is its marginal cost. ATC is total cost per member, so it falls as members join. Before class, use class 13's rule for a seller whose demand slopes down: ask how many members the space wants, where its price is read from, and how its profit appears in the figure.

Revealed in class.

When new spaces open: what happens to each one's demand?



Profit here is economic profit: every cost counted, the owner's own time included (class 11, slide 6).

Slide 8 When new spaces open: what happens to each one's demand?

The profit of the short run does not stay a secret: with free entry new spaces open, and each one is a close substitute for ours. In class we draw what that does to the demand our space faces; its cost curves are unchanged. Before class, ask whether this is a move along the old demand curve or a shift of it, what the space's rule now gives, and when new spaces stop opening.

Revealed in class.

In the long run: price, cost and spare desks

Revealed in class

Compare with class 12, slide 18: where does a price taker end up in the long run?
Reading pp. 270–271.

Slide 9 In the long run: price, cost and spare desks

Once entry has stopped, the figure shows where the space ends up: its price, its cost per member and how many of its 60 desks it fills. Before class, compare with class 12, slide 18: where does a price taker end up in the long run, in price against marginal cost and in price against the lowest ATC, and ask which of those two results a seller with a downward-sloping demand can share and which it cannot. Reading pp. 270–271.

Revealed in class.

Why do firms advertise?

Revealed in class

Reading pp. 272–276: the critique and the defence of advertising, and of brand names.
What is one more customer worth to a firm whose price is above its marginal cost?

Slide 10 Why do firms advertise?

The answer to slide 5's question, from the figure of slide 9. Before class, ask what one more customer is worth to a firm whose price is above its marginal cost, and what one more buyer is worth to a price taker selling at a price equal to marginal cost. Reading pp. 272–276: the critique and the defence of advertising, and of brand names.

Revealed in class.

In pairs, four minutes: a physiotherapy clinic

One of many clinics in town. Each session costs it 25 (the physio's pay, towels); the room's rent is 200 a week.

sessions a week	10	20	30	40	50
price a session, now (euros)	50	45	40	35	30
price after new clinics open	40	35	30	25	20

1. Its best number of sessions and price now?
2. Its profit a week now?
3. After new clinics open: sessions, price, profit?
4. In the long run, is its price above MC?

In pairs, four minutes: four answers.

MR per session: the extra revenue of ten more sessions ÷ 10 (class 13, slide 4).

Slide 11 In pairs, four minutes: a physiotherapy clinic

Exercise 1, worked in pairs: slides 7 to 9 on a table, with one of many clinics in town; each session costs it 25 euros and the room 200 a week. Before class, work out from the top two rows what each extra ten sessions adds to revenue per session, and compare that with what a session costs; ask what price goes with the number you choose, what is left after the rent, and how the last row changes all three answers.

Revealed in class.

The coworking space: which curve moves? (optional)

The coworking space after entry (slide 8)

1. It raises its price to 220 a month. Which curve moves?
2. A new space opens in the same street.
3. Its rent rises by 600 a month.

In pairs, two minutes: three answers.

A price change moves along the demand curve (class 5, slide 10).

Slide 12 The coworking space: which curve moves? (optional)

An optional exercise, in pairs, on what moves which curve, back at the coworking space after entry. Before class, ask for each event whether it changes the demand curve itself, one of the cost curves, or only the point on an unchanged curve, and which curves a fixed cost can touch.

Revealed in class.

Break

Ten-minute break

slide 13 of 23

Slide 13 Break

Ten minutes. After the break: four problems on Blocks 3 and 4, in pairs, the answer after each.

Problems on Blocks 3 and 4: in pairs, the answer after each

1. A tax on a market (classes 8–9)
2. A price taker's costs (classes 11–12)
3. An external cost (class 10)
4. A price cap (class 9)

slide 14 of 23

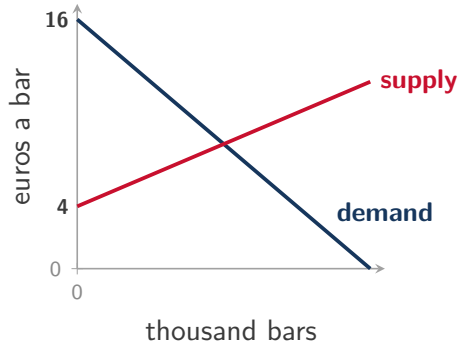
Slide 14 Problems on Blocks 3 and 4: in pairs, the answer after each

The second half: four problems from Blocks 3 and 4, each worked in pairs and answered right after, one move each with new numbers. Problems like these are good practice for Midterm 1 on Thursday 19 November.

In pairs, four minutes: a tax on turrón

A city's turrón in December: demand $P = 16 - Q/5$, supply $P = 4 + Q/10$ (Q thousand bars, P euros a bar).

1. The equilibrium, and the consumer surplus?
2. A tax of 3 euros a bar, paid by the sellers: which curve moves?
3. Bars sold; the price buyers pay; the price sellers keep?



In pairs, four minutes: three answers.

Consumer surplus: value minus price, the area under demand above the price (class 8, slide 6).

Slide 15 In pairs, four minutes: a tax on turrón

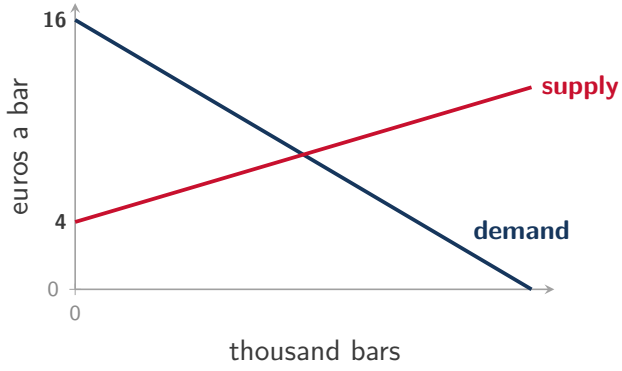
Problem 1, worked in pairs: a linear market with a per-unit tax, the move of classes 8 and 9; the figure shows the two curves of the city's turrón market in December. Before class, find the equilibrium by setting the two prices equal and the consumer surplus as a triangle; then ask which curve a tax paid by the sellers moves, in which direction and by how much, and how the price buyers pay and the price sellers keep are read from the figure afterwards.

Revealed in class.

Problem 1, continued: who bears the tax, and what is lost?

Same market, with the tax of 3 euros a bar.

4. Who bears more of the 3 euros, and why?
5. The tax revenue?
6. The deadweight loss?



In pairs, two minutes: three answers.

Deadweight loss: surplus lost on trades that no longer happen (class 8, slide 17).

Slide 16 Problem 1, continued: who bears the tax, and what is lost?

The same market, once the tax is in place. Before class, ask how to compare the price buyers pay and the price sellers keep with the old price, and why the side that responds less to a price change ends up with the larger share; then ask which quantity the revenue is counted on, and which trades the deadweight loss stands for.

Revealed in class.

In pairs, five minutes: a pomegranate farm

It sells crates at the market price. Its total cost a day:

crates a day	0	1	2	3	4	5	6
total cost (euros)	48	60	72	84	100	120	150
marginal cost							

1. The marginal cost of each crate?
2. At 28 euros a crate: how many crates, what profit?
3. Below which price does it harvest nothing (shut down for now)?
4. With free entry, the long-run price?

In pairs, five minutes: four answers.

Average variable cost (AVC): variable cost ÷ crates · ATC: total cost ÷ crates (class 11, slides 9–10).

Slide 17 In pairs, five minutes: a pomegranate farm

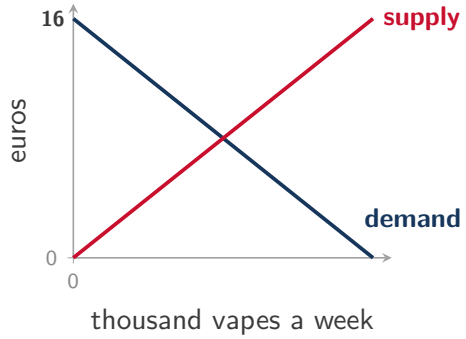
Problem 2, worked in pairs: a price taker's cost table, the move of classes 11 and 12. Before class, fill in from the total-cost row what each extra crate costs, and remember what the cost at zero crates is; then ask which crates are worth picking at a given price and what is left after all costs, below which price it is better to harvest nothing at all (shut down for now), and where free entry takes the price in the long run.

Revealed in class.

In pairs, four minutes: disposable vapes

A city's market: demand $P = 16 - Q/2$, supply $P = Q/2$ (Q thousand vapes a week, P euros). Each thrown-away vape's battery costs the city 2 euros to clean up.

1. How many vapes are sold, at what price?
2. The efficient number?
3. The tax that gets there; the loss without it?



In pairs, four minutes: three answers.

External cost: a cost that falls on people outside the deal (class 10, slide 6).

Slide 18 In pairs, four minutes: disposable vapes

Problem 3, worked in pairs: an external cost, the move of class 10, with invented numbers for the clean-up cost. Before class, find where the market settles on its own; then ask how the cost of one more vape to the whole city differs from the sellers' cost, where the quantity would be if that full cost were counted, what tax would get the market there, and which trades the loss without it stands for.

Revealed in class.

In pairs, three minutes: prawns at Christmas

A city's prawns in Christmas week (tonnes):

price (euros a kilo)	20	25	30	35
households want	60	50	40	30
caterers want	100	70	40	10
fishmongers offer	40	60	80	100

1. Market demand at each price?
2. The equilibrium?
3. A cap of 20 euros a kilo: tonnes sold; shortage?

In pairs, three minutes: three answers.

Market demand: add the buyers' quantities at each price (class 5, slide 11).

Price cap: a legal maximum price (class 9, slide 11).

Slide 19 In pairs, three minutes: prawns at Christmas

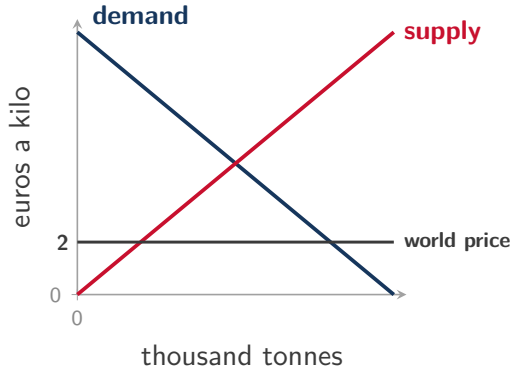
Problem 4, worked in pairs: a price cap on a market with two groups of buyers, the move of class 9. Before class, work out from the table how much both groups together want at each price, and ask what may be added across buyers and what may not; then find where the market clears, and ask what a legal maximum price below that does to the amount offered, the amount wanted and the amount actually sold.

Revealed in class.

A tariff on quinoa (optional)

A small country: home demand $Q = 100 - 10P$, home supply $Q = 10P$ (thousand tonnes; P euros a kilo). World price: 2 euros.

1. At the world price: home growers sell? Imports?
2. A tariff of 1 euro a kilo: imports; its revenue?
3. The surplus the country loses?



In pairs, three minutes: three answers.

Small country: it buys any amount at the world price (class 9, slide 18).

Slide 20 A tariff on quinoa (optional)

An optional problem, in pairs, on trade and a tariff, the move of class 9: a small country that buys quinoa at the world price. Before class, find how much is grown at home, how much is bought and how much is imported at that price; then ask what a tariff does to the home price, to the imports and to the state's revenue, and whose surplus the lost triangles come from.

Revealed in class.

Adverts: positive or normative? (optional)

1. "Betting adverts on TV raise how much young people bet."
2. "Betting adverts should be banned."
3. Two economists both want cheaper shopping. One would limit adverts; the other says adverts help new firms enter. What do they disagree about?

In pairs, two minutes: three answers.

Positive: a claim evidence can test · normative: a claim about what should be (class 4, slide 7).

Slide 21 Adverts: positive or normative? (optional)

An optional exercise, in pairs, on positive and normative claims, with adverts as the subject. Before class, ask for each claim whether evidence could test it or whether it states what should be, and what two people who share a goal but give opposite advice are really disagreeing about.

Revealed in class.

Monopolistic competition: many sellers, each a little different, free entry.

Short run: like a monopoly.
Long run: entry until profit is zero.

Even then the price stays above MC, and capacity is left unused.

Slide 22 In one line

The class in three lines, to read slowly: the first is slide 6, the second slides 7 and 8, the third slide 9. A fourth line, which answers slide 5's question, is revealed in class.

Before Monday 16 November

- Monday 16 November: review for Midterm 1 (Blocks 1–4), the full two hours
- Thursday 19 November: Midterm 1 (Blocks 1–4), in class
- Practise in Sextant: the units the course page lists



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Slide 23 Before Monday 16 November

Monday 16 November: the review session for Midterm 1, Blocks 1 to 4, the whole class. Thursday 19 November: Midterm 1, Blocks 1 to 4, in class. Problems like today's four are good practice; Sextant has more, in the units the course page lists.

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